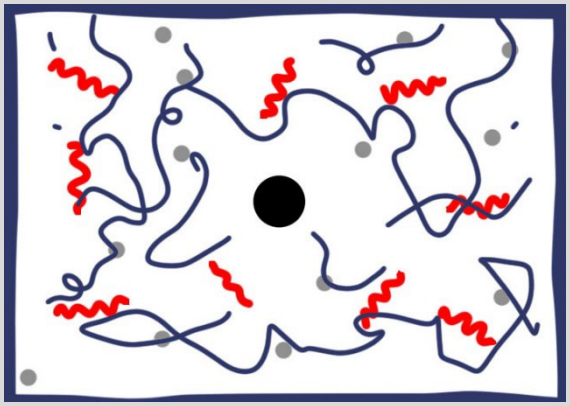
Approaching Dirty Black Holes

Exact anisotropic models, test field propagation, and environmental effects



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UIUC Physics, ICASU, ARC 2026

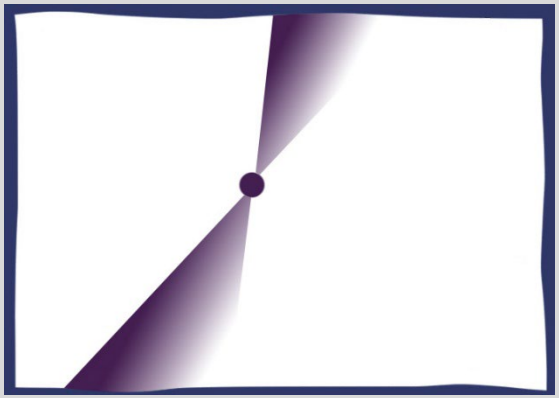
Roadmap

- **Why study dirty black holes?**
- **A simple exact model**
- **From geometry to observables**
- **Toward time-domain evolution**

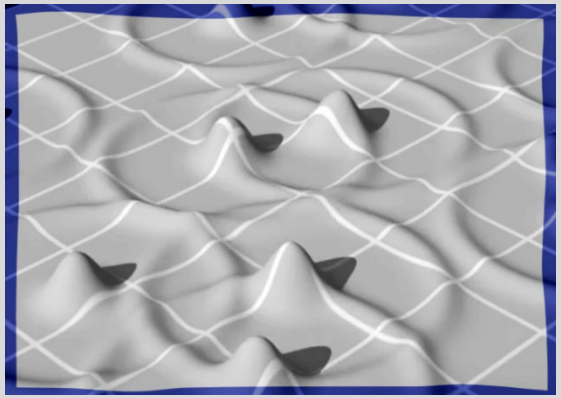
**This work was carried out in collaboration with the Witek Gravity Group:
Helvi Witek, Domenica Garzon, Yi Zhang, Zeyu Zeng.**

Why care about non-vacuum black holes?

Gravitational waves
ringdown spectroscopy



Precision astrophysics
timing / imaging



Precision cosmology
environmental systematics

Need to distinguish intrinsic black hole physics from environmental effects.

How do we model black holes in matter?

Black hole + matter exact solutions

Analytic control interpretable parameters

Kiselev-type solutions,
Lemaitre-Tolman-Bondi

Effective exterior models

Flexibility, broad parameter space

Burkert, Moore, NFW
dark matter halo models

Numerical, matched solutions

Full-system modeling, realism

Einstein-Straus,
Cosmological NR

How do we model black holes in matter?

Black hole + matter exact solutions	Effective exterior models	Numerical, matched solutions
Analytic control interpretable parameters	**Flexibility, broad parameter space**	**Full-system modeling, realism**
Kiselev-type solutions, Lemaitre-Tolman-Bondi	Burkert, Moore, dark matter halo models	Einstein-Straus, Cosmological NR

This talk: exact solutions as controlled, interpretable baselines.

Why exact solutions

- fully relativistic, no approximation in the background
- interpretable parameters with clear physical meaning
- exact horizon structure available analytically
- controlled perturbation theory on a known background
- clean benchmarks for numerical codes and analytic approximation schemes

metric → horizons → effective potential → observables

Static black holes surrounded by anisotropic matter

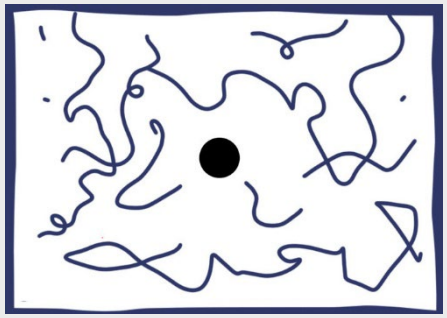
$$\mathrm{d}s^2 = -f(r)\mathrm{d}t^2 + f(r)^{-1}\mathrm{d}r^2 + r^2\mathrm{d}\Omega^2$$

$f(r)$ encodes both the black hole and its environment

Energy-momentum tensor $T_\mu^\nu = \mathrm{diag}(-\varepsilon, p_r, p_\perp, p_\perp)$

Matter parameters

- **energy density $\varepsilon(r)$**
- **radial pressure $p_r(r)$**
- **tangential pressure $p_\perp(r)$**
- **with $p_r \neq p_\perp$ in general**



A simple exact laboratory: Kiselev-type black holes

$$T^{\mu}_{\nu} = \sum_i T^{\mu}_{\nu(i)}$$

mixture of non-interacting sources

$$\frac{p_{r(i)} + 2p_{\perp(i)}}{3} = w_i \varepsilon_i$$

effective barotropic state equation

$$f(r) = 1 - \frac{2M}{r} - \sum_i \frac{K_i}{r^{3w_i+1}}$$

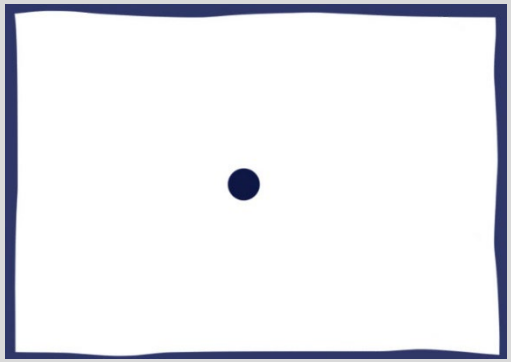
as many terms as you want

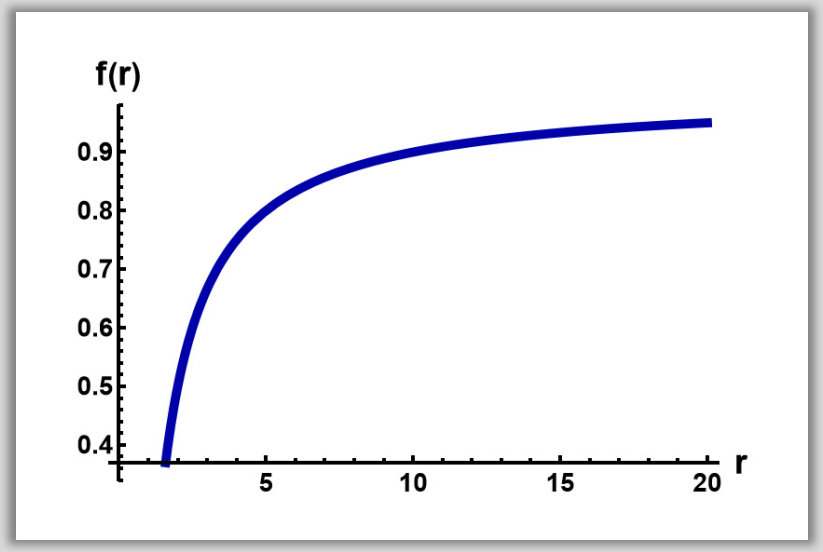
metric → horizons

Number of horizons ≈ number of sources + 1

Zeng, Kopteva, arXiv:2603.21040

Kiselev particular case: Schwarzschild

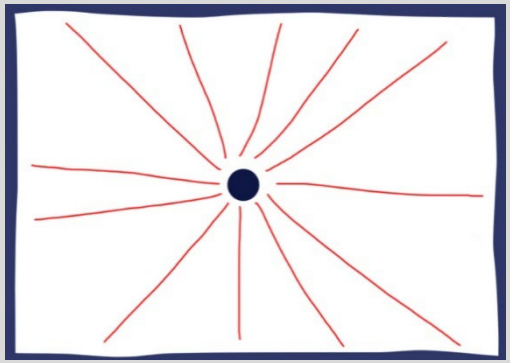


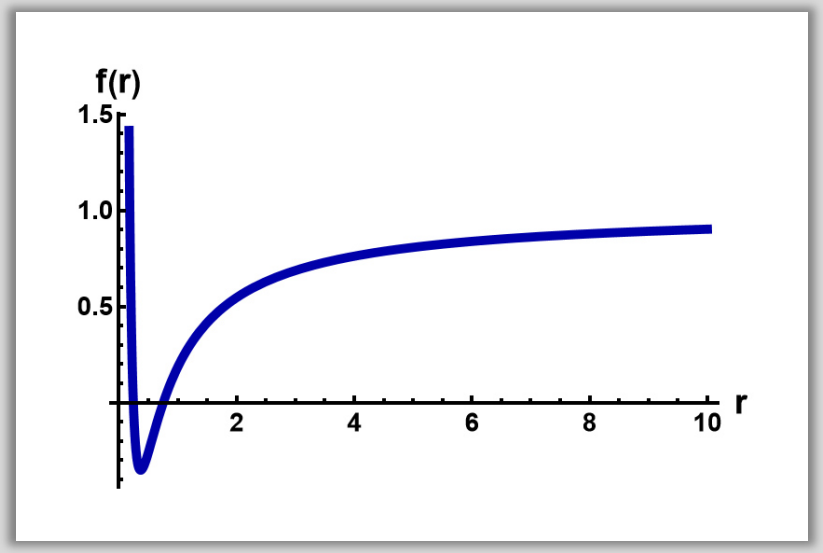


$$f(r) = 1 - \frac{2M}{r}$$

$$K_i = 0, \;\; w_i = 0$$

Kiselev particular case: Reissner-Nordstrom

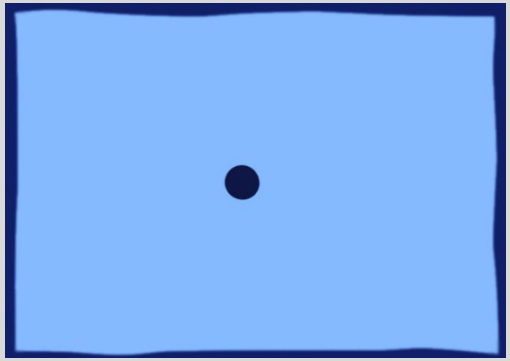


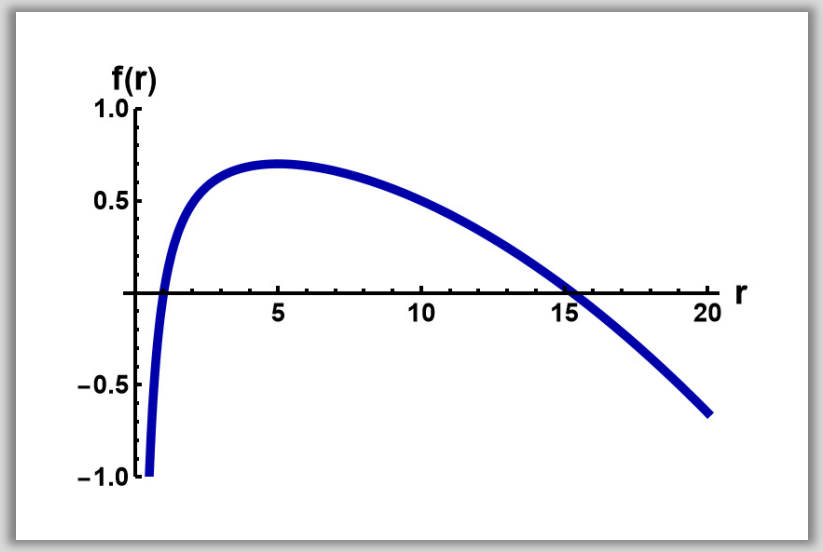


$$f(r) = 1 - \frac{2M}{r} + \frac{Q^2}{r^2}$$

$$K_1 = -Q^2, \; w_1 = 1/3$$

Kiselev particular case: Kottler

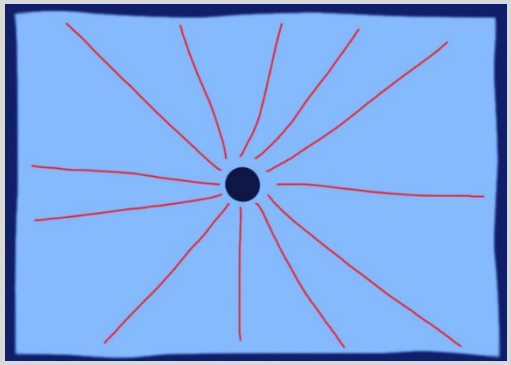


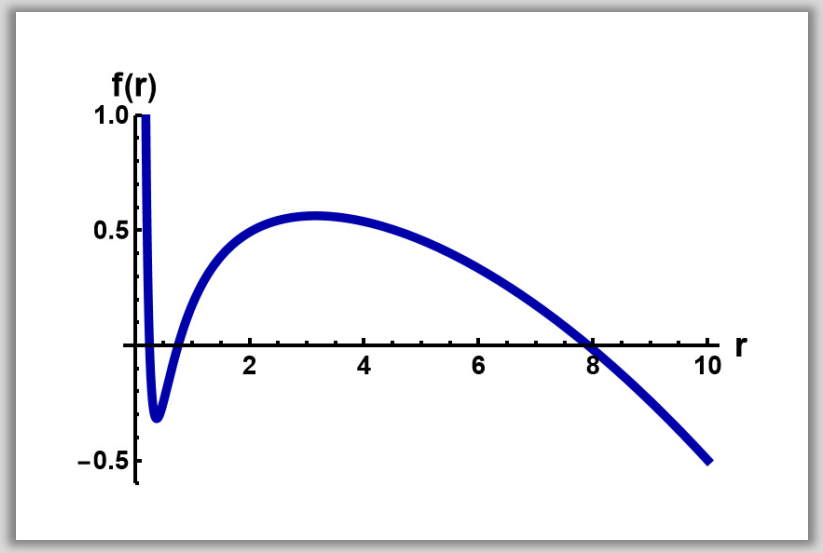


$$f(r) = 1 - \frac{2M}{r} - \frac{\Lambda}{3} r^2$$

$$K_1 = \Lambda/3, \;\; w_1 = -1$$

Kiselev particular case: RNdS

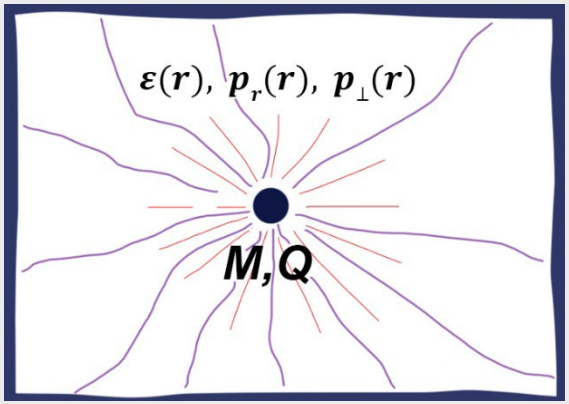


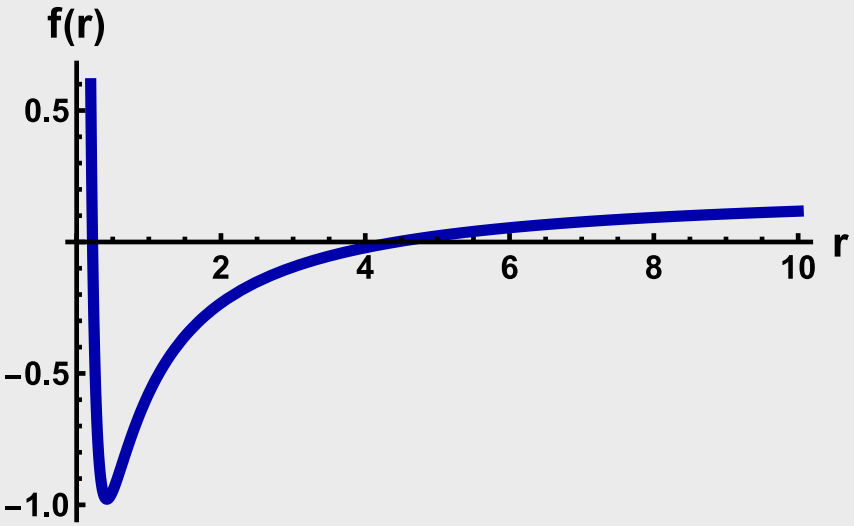


$$f(r) = 1 - \frac{2M}{r} + \frac{Q^2}{r^2} - \frac{\Lambda}{3} r^2$$

$K_1 = -Q^2$, $w_1 = 1/3$, $K_2 = \Lambda/3$, $w_2 = -1$

Kiselev particular case: string-like fluid

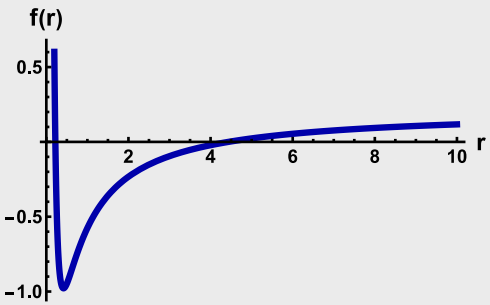




$$f(r) = 1 - K - \frac{2M}{r} + \frac{Q^2}{r^2}$$

$K_1 = K,\ w_1 = -1/3,\ \ K_2 = -Q^2,\ w_2 = 1/3$

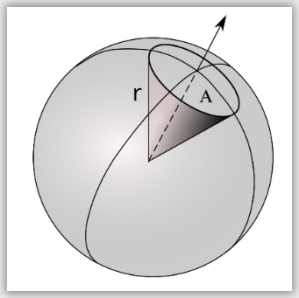
Black hole immersed in string-like fluid



$$f(r) = 1 - K - \frac{2M}{r} + \frac{Q^2}{r^2}$$

$$f(\infty) = 1 - K$$

$$\mathrm{d}l^2 = (1-K)^{-1}\mathrm{d}r^2 + r^2\mathrm{d}\Omega^2$$



$$\varepsilon_1 = \frac{K}{r^2}, \quad p_r = -\frac{K}{r^2}, \quad p_\perp = 0 \qquad \varepsilon_2 = \frac{Q^2}{r^4}, \quad p_r = -p_\perp = -\frac{Q^2}{r^4}$$

Quasi-asymptotically flat spacetime with **solid angle deficit** and globally conical asymptotic, similar to the Letelier string cloud and global monopole geometries

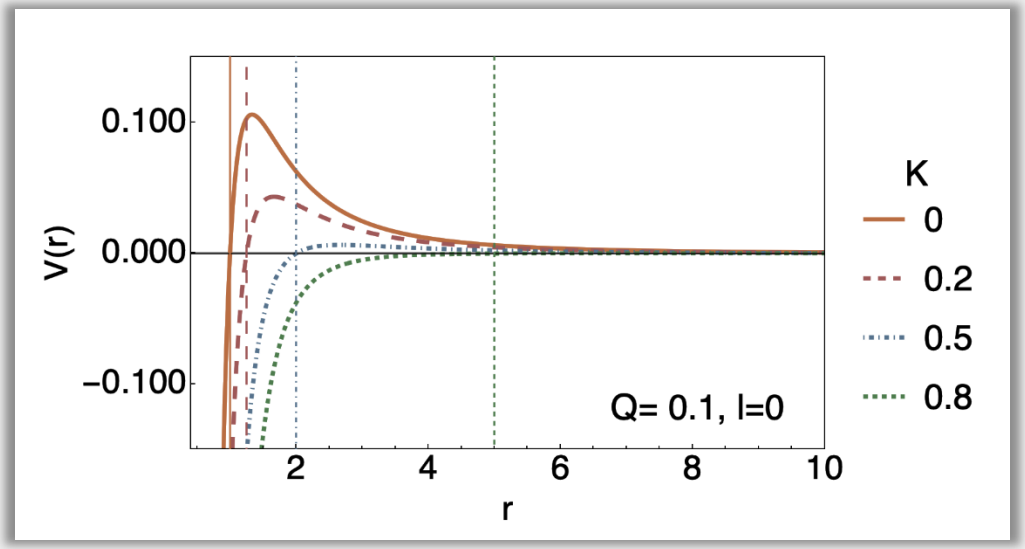
Parameter bounds come from physically reasonable matter content, and the existence of regular static exterior with time-like time: $0 \leq K < 1 \quad Q < \mathrm{M}$

From geometry to observables

Having background geometry

- → **put a test field on it**
- → **reduce to the radial equation**
- → **read off the effective potential**
- → **study quasi-normal modes**

charged black hole M,Q + anisotropic matter K
massless charged scalar field q



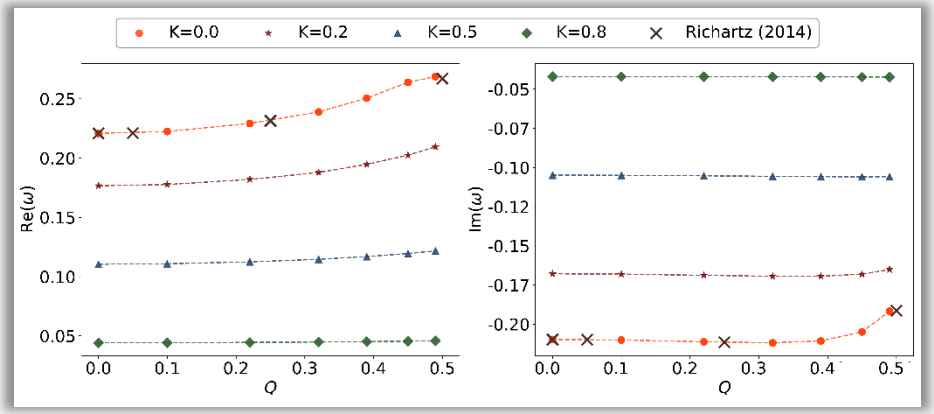
Charged scalar perturbations and quasinormal spectrum

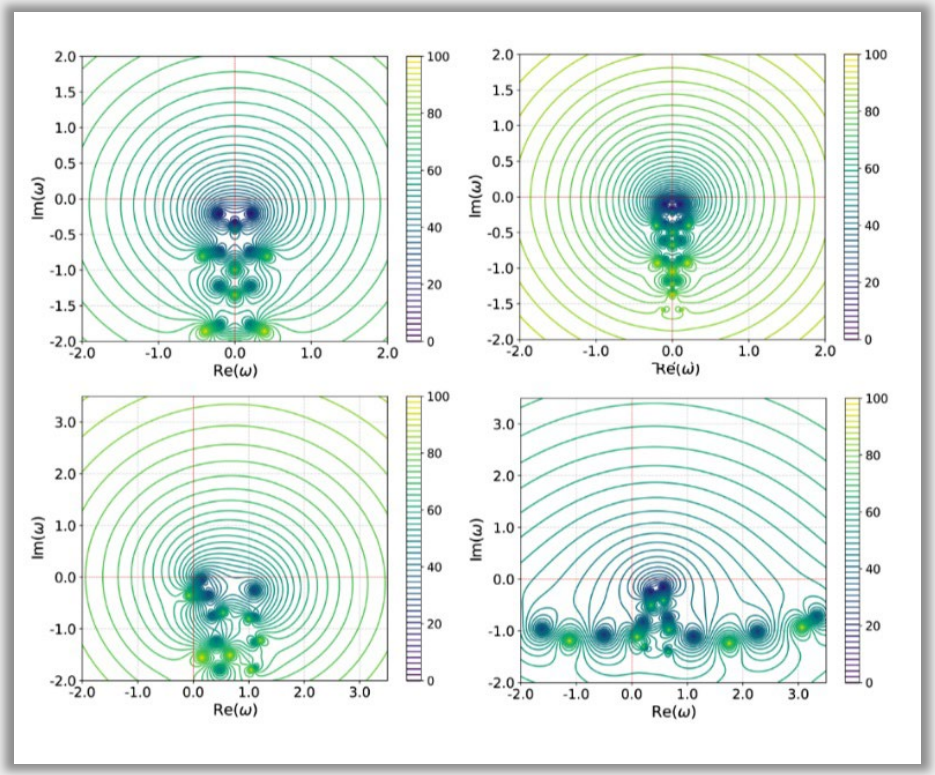
$$(\nabla_\mu - iqA_\mu)(\nabla^\mu - iqA^\mu)\Phi = 0$$

$$\frac{d^2\Psi(r)}{dr_*^2} + \left[\left(\omega - \frac{Qq}{r}\right)^2 - V(r)\right]\Psi(r) = 0$$

$$\frac{\mathrm{d}r_*}{\mathrm{d}r} = \frac{1}{f(r)}$$

$$V(r) = f(r)\left(\frac{l(l+1)}{r^2} + \frac{f'(r)}{r}\right)$$





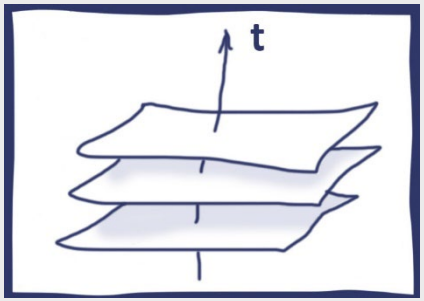
Methods:

Leaver's continued fraction, WKB

by Domenica and Yi

Toward time-domain evolution: isotropic coordinates

$$r \longrightarrow \rho$$



- Isotropic coordinate map ρ(r) derived as a convergent series for the full Kiselev family with any number of sources
- Inversion r(ρ) obtained via Lagrange-Bürmann theorem
- Metric cast in 3+1-ready form:

$$\mathrm{d}s^2 = -F(\rho)\mathrm{d}t^2 + \Phi(\rho)^4\left(\mathrm{d}\rho^2 + \rho^2\mathrm{d}\Omega^2\right)$$

$$F(\rho) = f(r(\rho)),\ \Phi(\rho)^2 = r(\rho)/\rho$$

- Covers all known limits: Schwarzschild, RN, Kottler, RNdS

Zeng, Kopteva, arXiv:2603.21040

- Target: RN + single Kiselev term as controlled, interpretable background
- Test field: charged massive scalar
- Implementation in Einstein Toolkit in progress
- Goal: time-domain ringdown wave-forms with explicit dependence on matter parameter K

Summary and outlook

Summary:

- **Black holes in matter require controlled non-vacuum baselines.**
- **Kiselev-type exact solutions provide clear and interpretable laboratory for this.**
- **Our work connects exact geometry, perturbation theory, and numerical implementation within an analytically controlled framework.**

Outlook:

- **Kiselev family is rich: multi-source configurations open a path toward modelling anisotropic stars and halos.**

Thank you

Backup A: parameter space

$$\rho = -p_r = -\sum_{i=1}^{n} \frac{3K_i w_i}{r^{3(w_i+1)}}, \qquad p_\perp = -\sum_{i=1}^{n} \frac{3K_i w_i (3w_i+1)}{2r^{3(w_i+1)}}$$

DEC $\rho \geq 0,\ \rho \geq |p_\perp|$.

I. $\boldsymbol{-1 \leq w < 0}$ **and** $\boldsymbol{K > 0}$

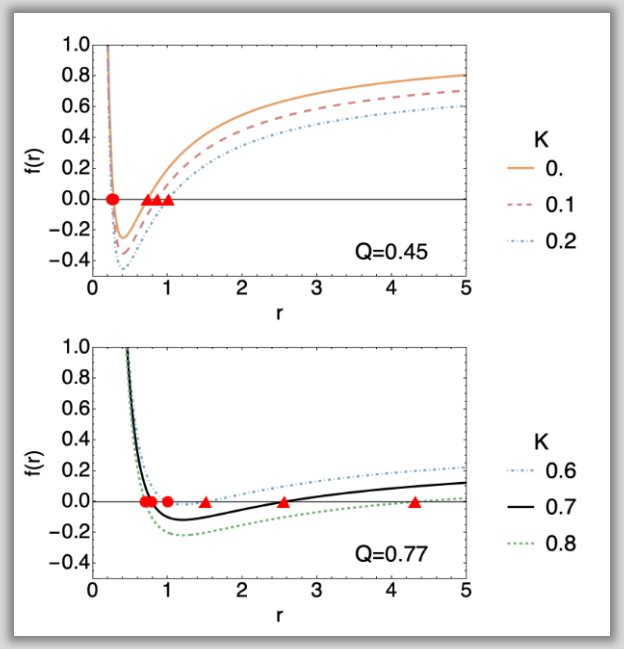
II. $\boldsymbol{0 < w \leq 1/3}$ **and** $\boldsymbol{K < 0}$

III. $\boldsymbol{w > 1/3}$ **and** $\boldsymbol{K > 0}$

$$\boldsymbol{r \geq r_{\mathrm{DEC}} := \left(\frac{9Kw(w+1)}{4Q^2}\right)^{\frac{1}{3w-1}}}$$

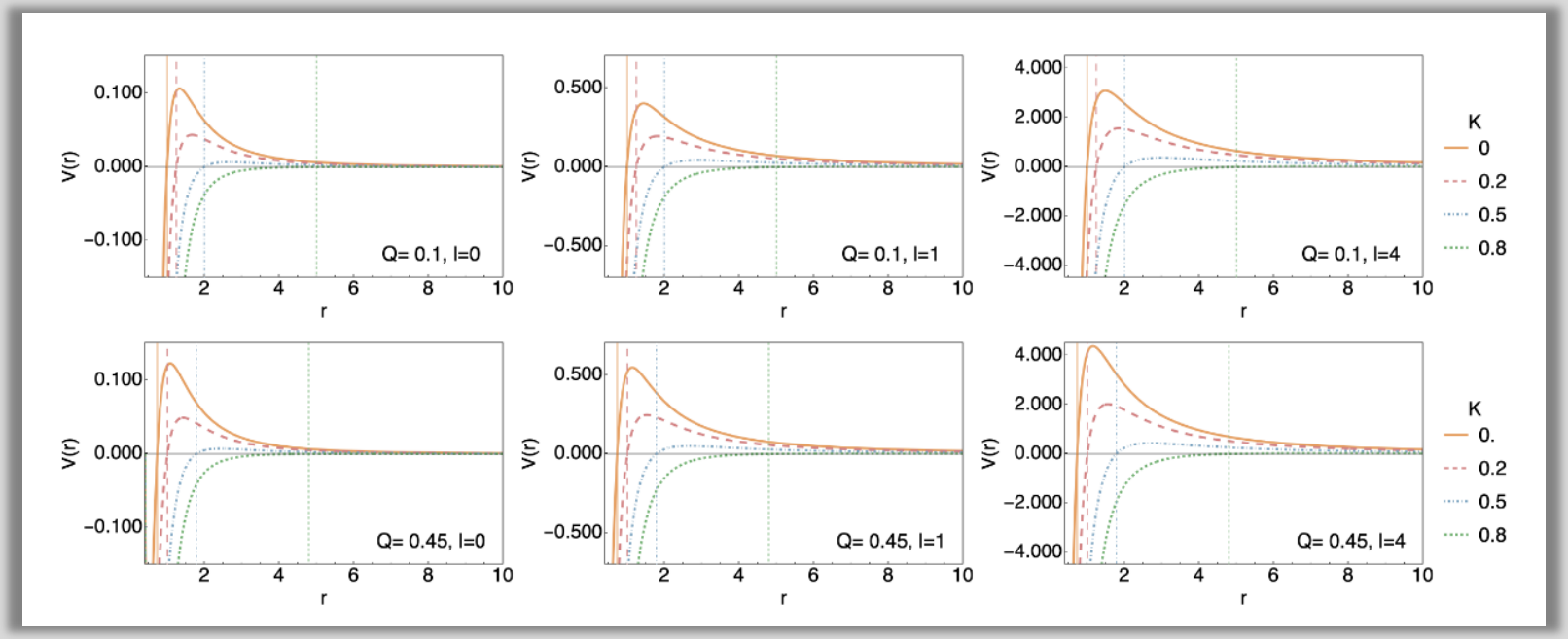
Backup B: horizons

$$r_{\pm} = \frac{1}{2(1-K)}\left(1 \pm \sqrt{1-4(1-K)Q^2}\right)$$



Backup C: effective potential

$$V(r) = -\frac{2Q^4}{r^6} + \frac{3Q^2}{r^5} + \frac{\lambda Q^2 - 2(1-K)Q^2 - 1}{r^4} + \frac{1-K-\lambda}{r^3} + \frac{\lambda(1-K)}{r^2}$$



Backup D: Curvature to Isotropic coordinates map

$$ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\sigma^2 \qquad ds^2 = -F(\rho)dt^2 + \Phi(\rho)^4(d\rho^2 + \rho^2 d\sigma^2)$$

$$\rho(r) = \rho_0\, r\, e^{I(r)} \qquad I(r) = \int_{r_+}^{r} \frac{dx}{x}\left(\frac{1}{\sqrt{f(x)}} - 1\right)$$

$$I(r) = \sum_{|\mathbf{m}|=k,\, k=1} C(\mathbf{m})\; r_g^{m_0} \prod_{\mathrm{i}=1}^{n} K_{\mathrm{i}}^{m_{\mathrm{i}}} \int_{r_+}^{r} dx\, x^{-1-\alpha(\mathbf{m})}$$

$$C(\mathbf{m}) = \frac{(2|\mathbf{m}|)!}{4^{|\mathbf{m}|}(|\mathbf{m}|!)^2} \frac{|\mathbf{m}|!}{\prod_{\mathrm{i}=0}^{n}(m_{\mathrm{i}}!)} \qquad \alpha(\mathbf{m}) = m_0 + \sum_{\mathrm{i}=1}^{n} m_{\mathrm{i}} s_{\mathrm{i}}$$

Backup E: Lagrange-Burmann inversion

$$\rho_\star \coloneqq \rho(r_\star) \qquad \Psi(r) \coloneqq \rho(r)$$

$$r(\rho) = r_\star + \sum_{n=1}^{\infty} \frac{(\rho-\rho_\star)^n}{n!} \left[\frac{\mathrm{d}^{\,n-1}}{\mathrm{d}r^{\,n-1}} \left(\frac{r-r_\star}{\Psi(r)-\Psi(r_\star)} \right)^n \right]_{r=r_\star} = r_\star + \sum_{n=1}^{\infty} c_n (\rho-\rho_\star)^n$$

Converges in the open disk: $|\rho-\rho_\star| < R, \qquad R = \mathrm{dist}\big(\rho_\star, \Sigma_\rho\big)$